# FLOW CHART
# AND
# DESIGN CHOICE DOCUMENT

## GROUP D8


## GROUP MEMBERS

E/21/389 Srikanthan S.
E/21/390 Sriman K.
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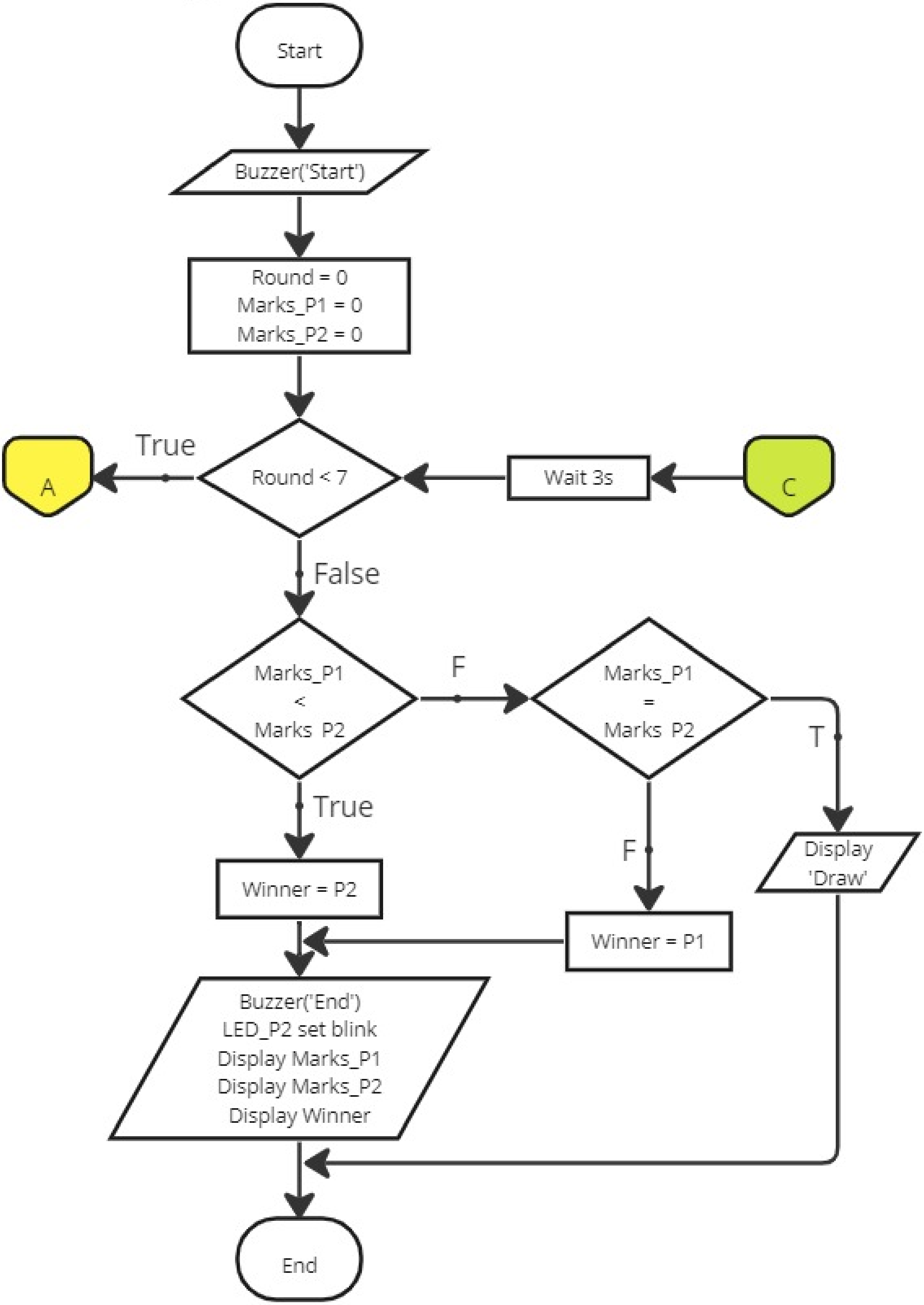
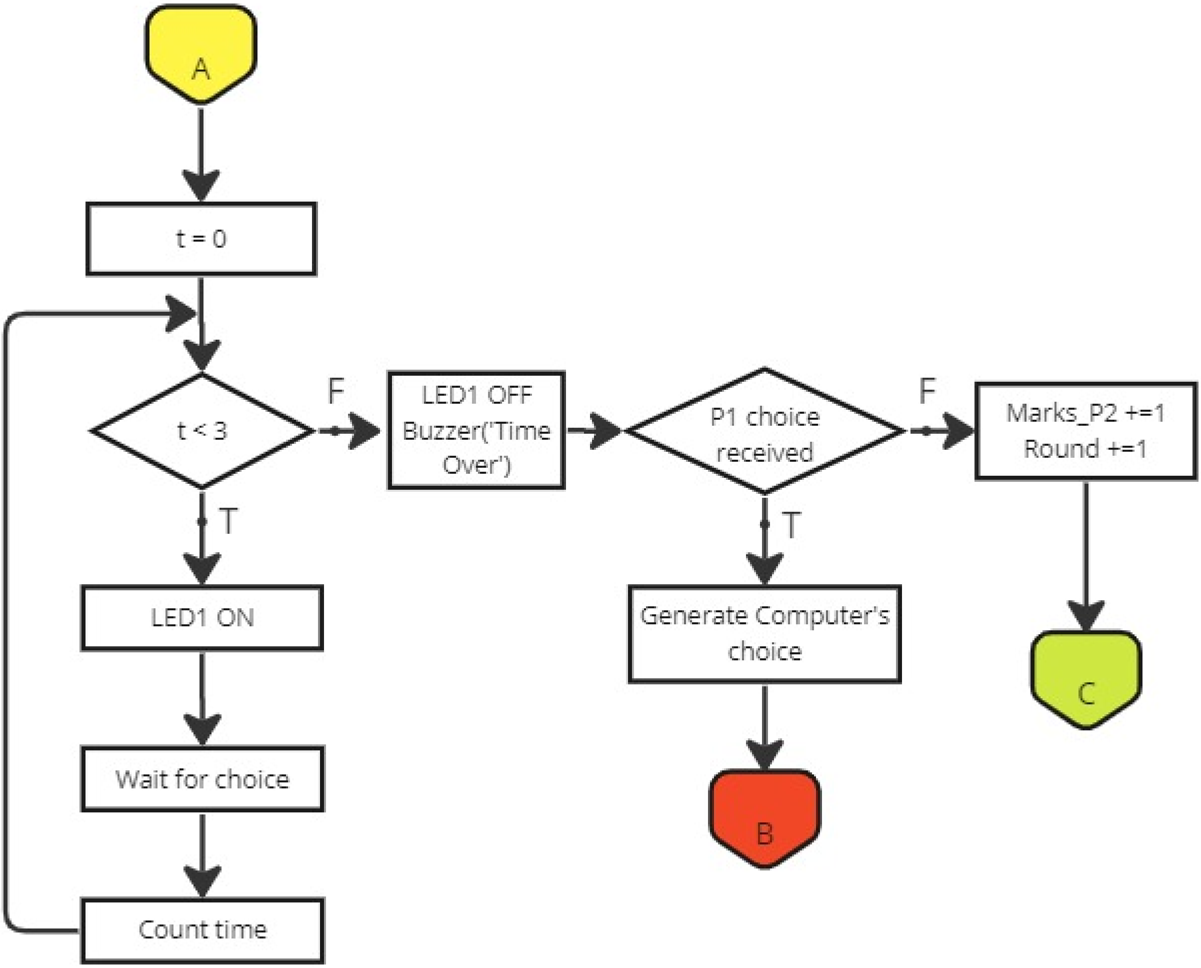
# Main Program

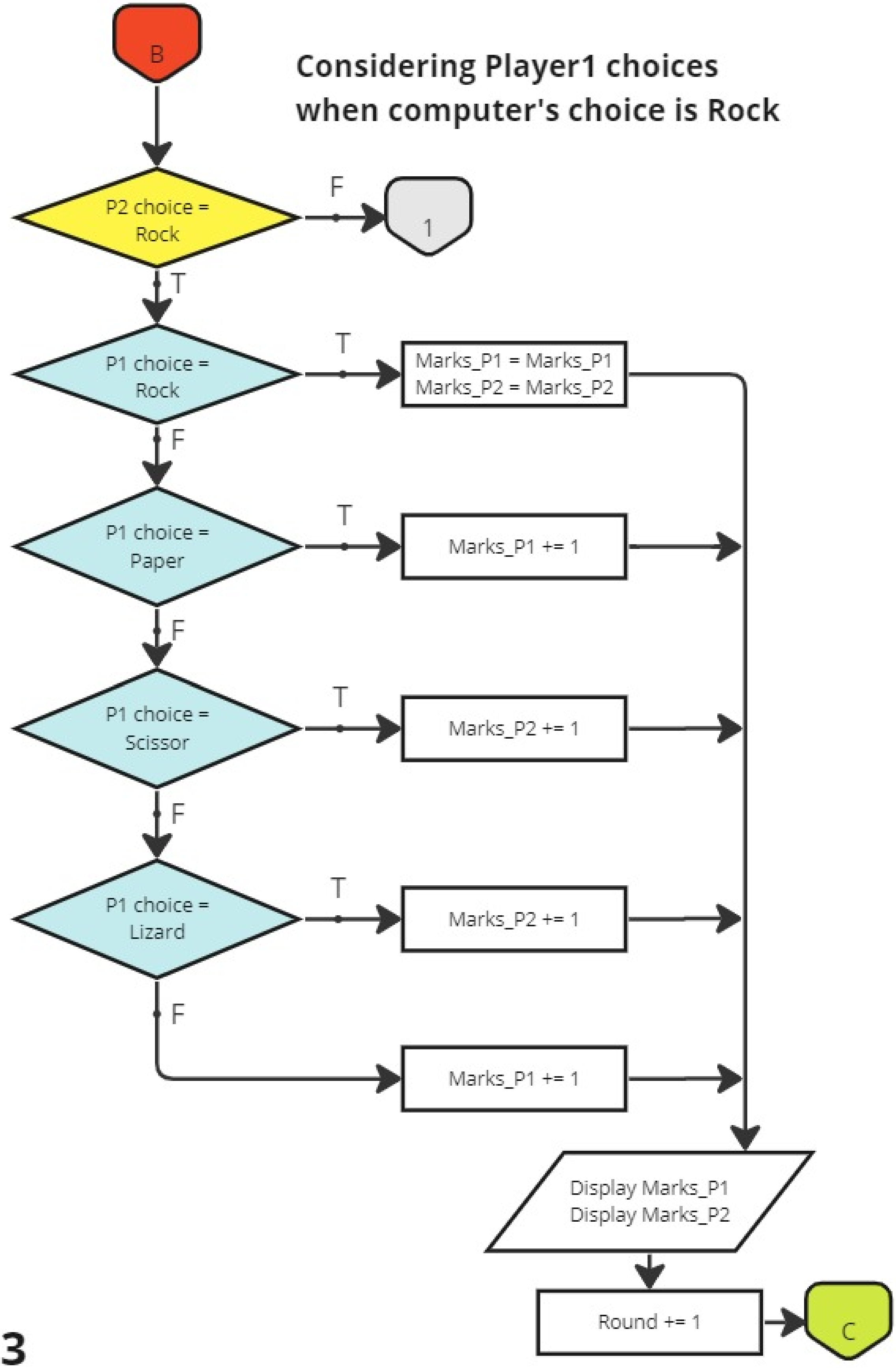



# No 1

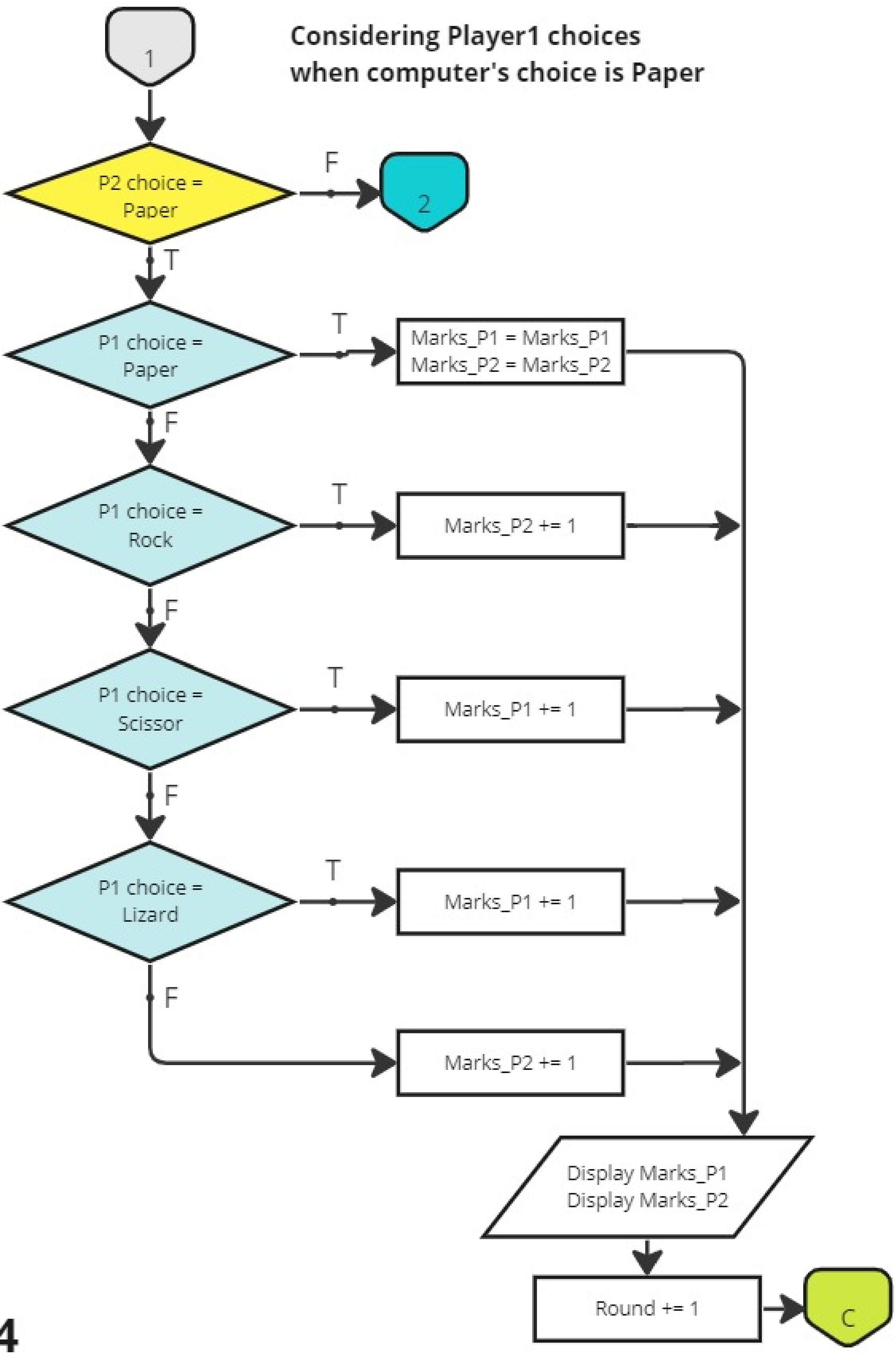
# Taking Player1 choice within the given time




# No 2

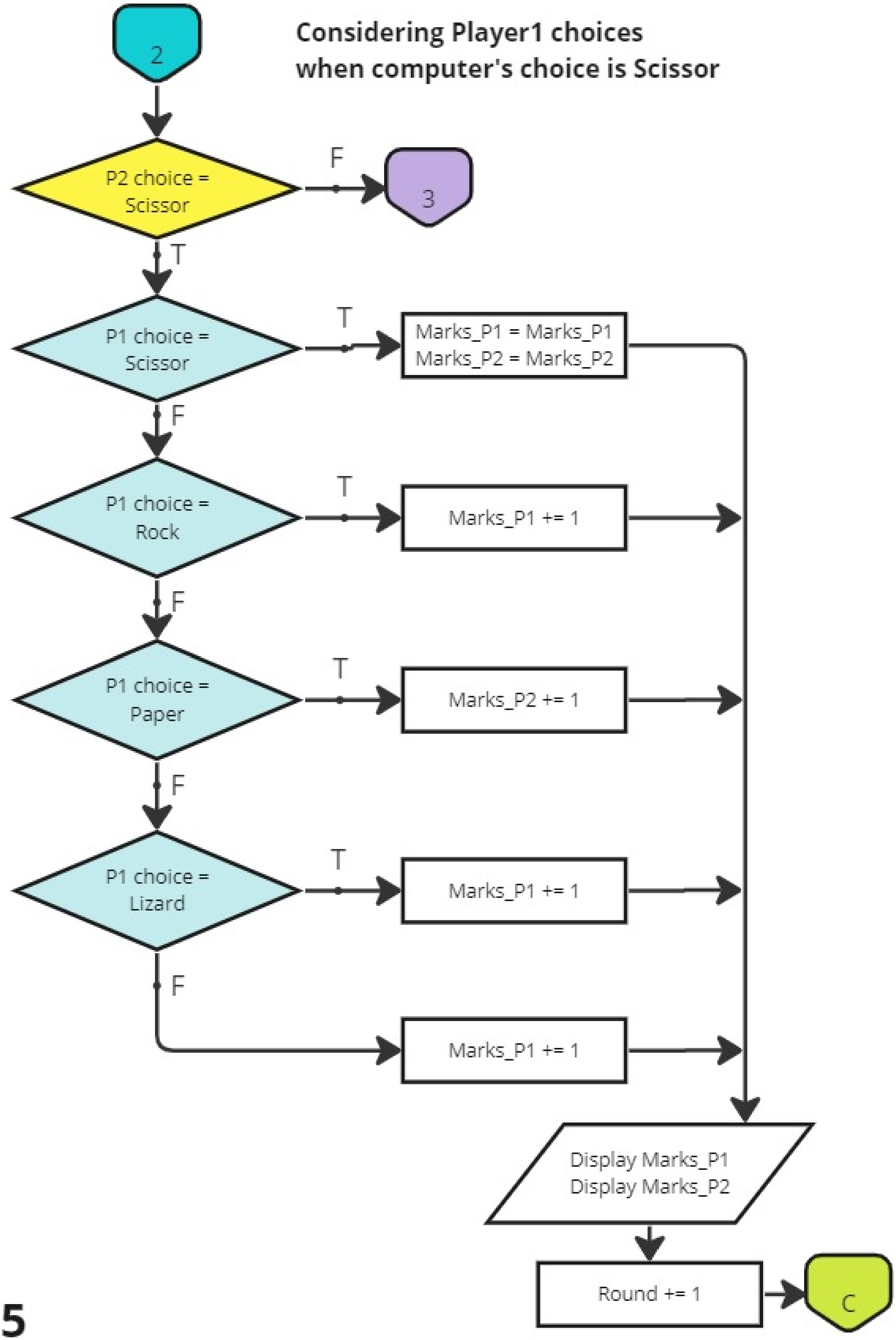

B
Considering Player1 choices
when computer's choice is Rock
P2 choice =
Rock
F
1
T
P1 choice =
Rock
T
Marks_P1 = Marks_P1
Marks_P2 = Marks_P2
F
P1 choice =
Paper
T
Marks_P1 += 1
F
P1 choice =
Scissor
T
Marks_P2 += 1
F
P1 choice =
Lizard
T
Marks_P2 += 1
F
Marks_P1 += 1
Display Marks_P1
Display Marks_P2
Round += 1
C

No 3


Considering Player1 choices
when computer's choice is Paper
1
P2 choice = Paper
F
2
T
P1 choice = Paper
T
Marks_P1 = Marks_P1
Marks_P2 = Marks_P2
F
P1 choice = Rock
T
Marks_P2 += 1
F
P1 choice = Scissor
T
Marks_P1 += 1
F
P1 choice = Lizard
T
Marks_P1 += 1
F
Marks_P2 += 1
Display Marks_P1
Display Marks_P2
Round += 1
C

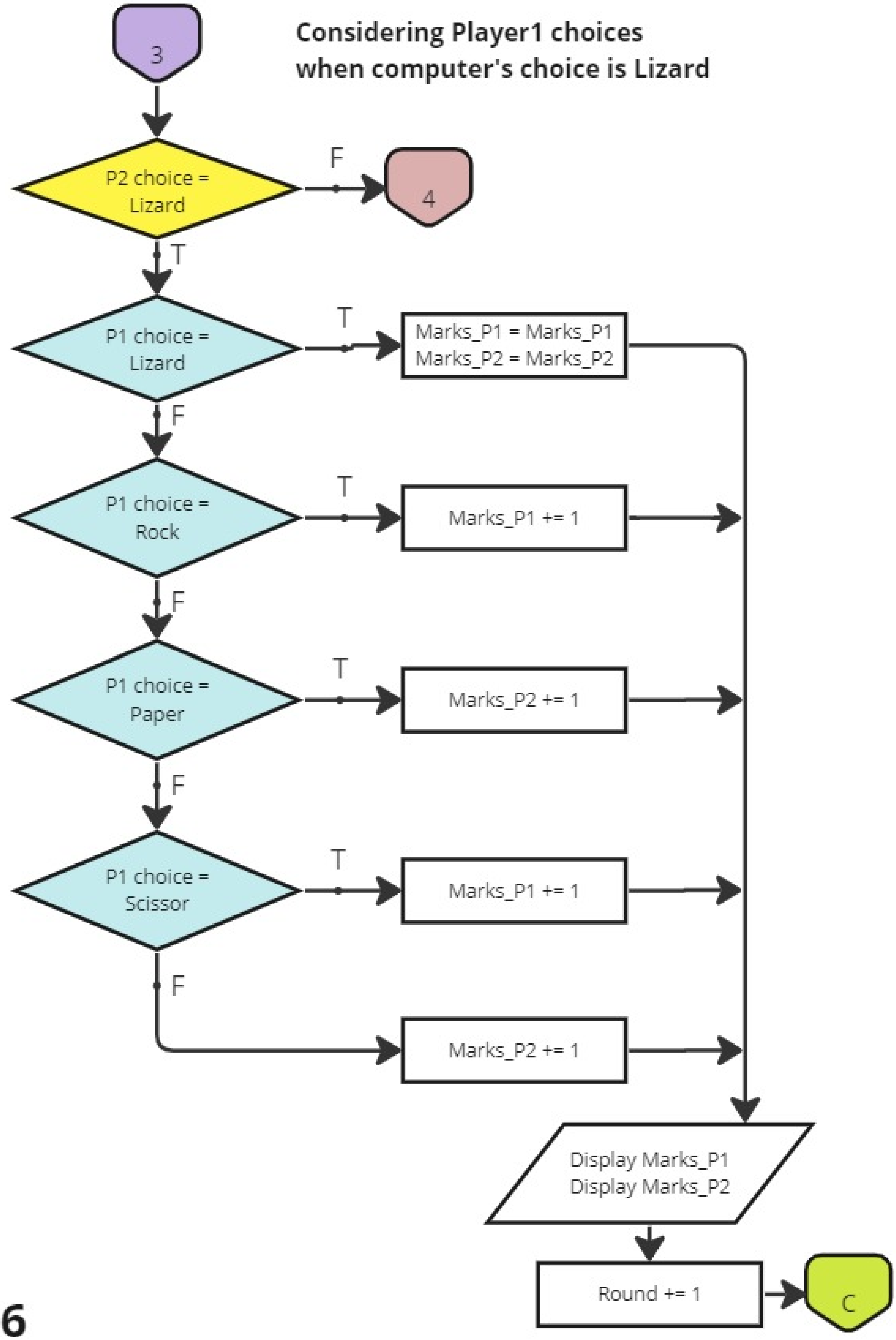
**No 4**

Considering Player1 choices
when computer's choice is Scissor



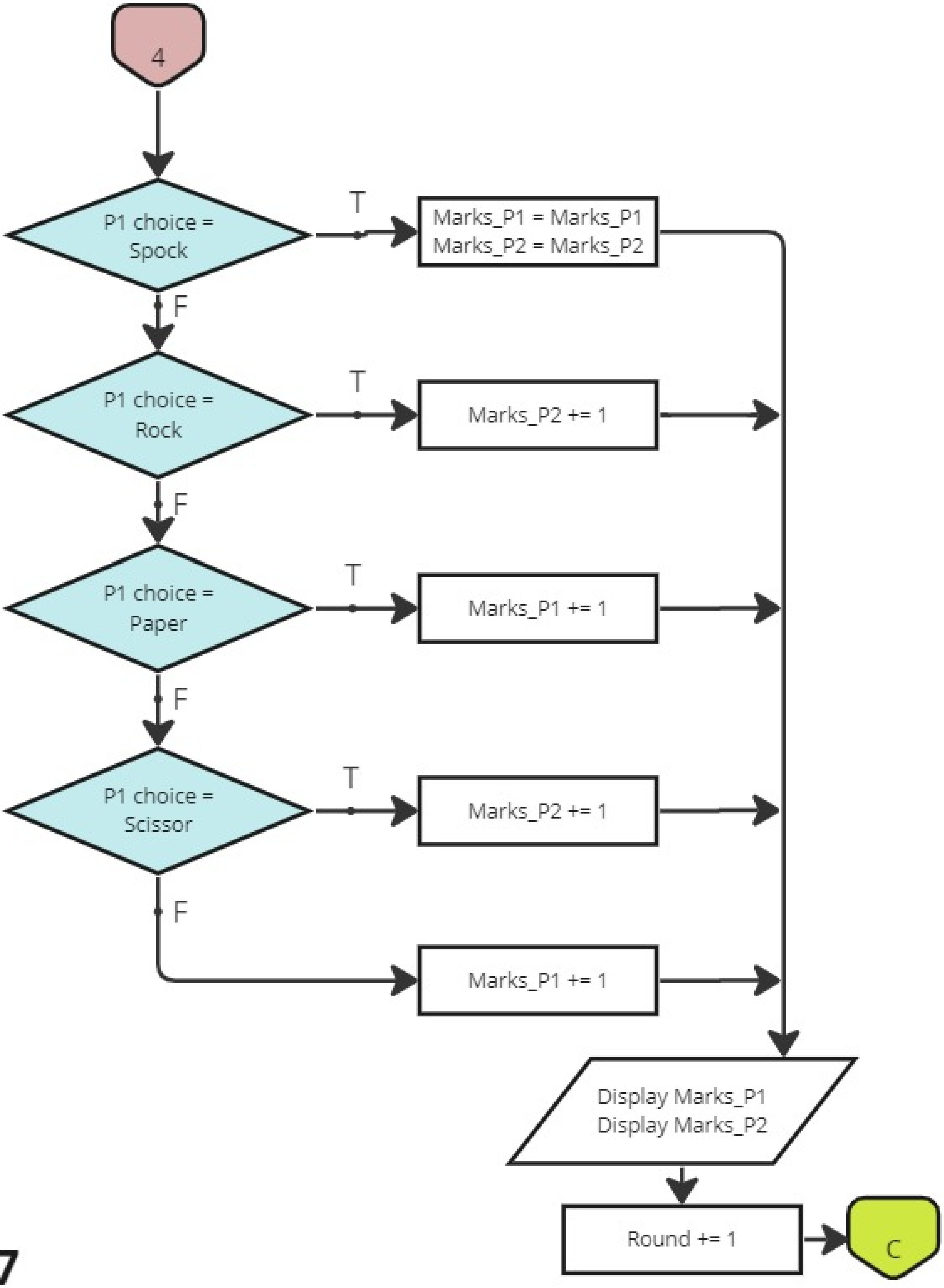

**No 5**

Considering Player1 choices when computer's choice is Lizard




No 6

**Considering Player1 choices when computer's choice is Spock**




**No 7**

**Rock Paper Scissor Lizard Spock**

Group - D8

Group members

| | |
|---|---|
| E/21/389 | Srikanthan S. |
| E/21/390 | Sriman K. |
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**Decisions**

We decided to divide the flowchart into a few parts. They are,

1. **The main flow chart** - including the start and end of the entire flow chart, controlling the number of rounds and ultimately displaying the final results of the game.
2. **Time checker** - this checks whether the human player (P1) gives a choice within the allowed time.
3. **Comparing Player1 choices with computer choices**

This made our flowchart much easier to understand and present.

**Variables**

Round = number of rounds
Marks_P1 = marks of player1 (human)
Marks_P2 = marks of player2 (computer)
Winner = winner of the game
t = time in seconds

**Other notations**

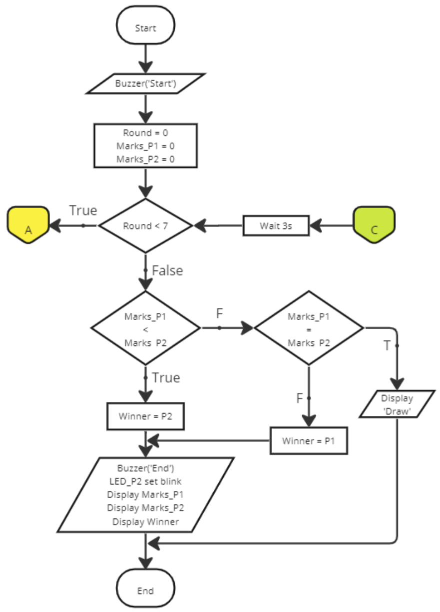
LED1 - Indicating LED

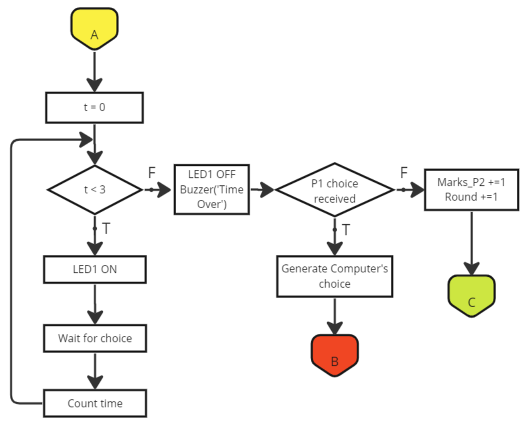
**Function of the flow chart**

**Flowchart No1**

- Flow chart starts
- Buzzer buzzes ‘Start’ tone
- Set the values Round = 0, Marks_P1 = 0 and Marks_P2 = 0
- Checks if the number of rounds is less than 7 or not
    - **If true -** time is counted upto 3 seconds and if Player1 gives a choice within 3 seconds, computer’s choice is generated and it is compared with Player1’s choice. (Assumption: computer’s choice is independent of Player1’s choice.) If Player1 fails to give a choice within 3 seconds, Player2 will get one mark, the indicator LED turns off, buzzer buzzes “Time Over” tone and number of rounds is incremented by 1. Then connects with main flow chart to check the number of rounds completed.

    - **If false -** the current total marks of players are compared and the winner is selected. Then, the buzzer buzzes ‘End’ tone, LED set corresponding to computer blinks and displays the marks of each player and the winner. Then flowchart ends.




## Comparing the choices of Player1 and computer

In each round if Player1 gives a choice within the given 3 seconds the computer's choice is generated and then,

1. First, it is checked if computer's choice is 'Rock'. If it is true then the choices of Player1 are checked. If false, flow chart connects with flowchart no. 4,5,6,7 to check computer's choice for other options : 'Paper', 'Scissor', 'Lizard', 'Spock'.
2. If there is a tie the marks do not change.
3. According to the different possible choice combinations, if Player1 wins, he gets one mark or if computer wins, he gets one mark.
4. After all the possible choices from Player1 are checked marks of both players are displayed and the number of rounds is incremented by 1. Then connects with main flow chart through off page connector 'C'.
5. Likewise, all the possible combinations of choices are considered and marks are given.

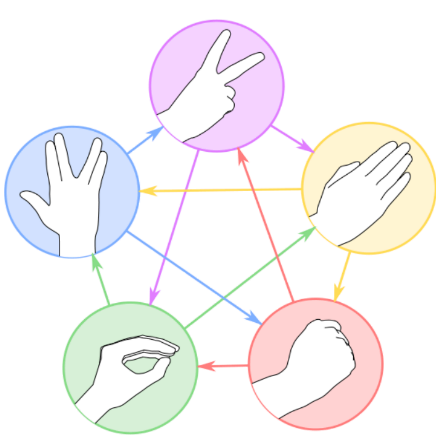
**Terminating rounds**

While the number of rounds is less than 7 the rounds are iterated and when the number of rounds becomes 7 it means game has completed 7 rounds. When flow chart connects with main flowchart with number of counts as 7 through off page connector ‘C’, the winner is selected and the game ends.

**Marking**

Table1

| **Choice1** | **Choice2** | **Winner** |
|---|---|---|
| Scissor | Paper | Scissor |
| Paper | Rock | Paper |
| Rock | Lizard | Rock |
| Lizard | Spock | Lizard |
| Spock | Scissor | Spock |
| Scissor | Lizard | Scissor |
| Lizard | Paper | Lizard |
| Paper | Spock | Paper |
| Spock | Rock | Spock |
| Rock | Scissor | Rock |



**How to play…**

Press Button_1 to start game

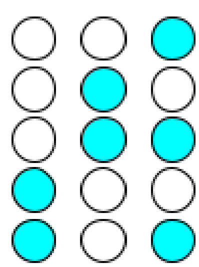
Start/End

Buzzer buzzes start tone
Indicator LED turns on
Player 1 should give choice using push buttons within 3 seconds

Rock Paper Scissor Lizard Spock

After 3 seconds, indicator LED turns off and buzzer buzzes time over.
If player did not give a choice computer get a mark.
Otherwise, computer generates a choice and it is displayed by LEDs.

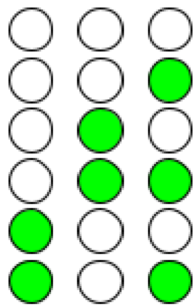
| | |
|---|---|
| Rock | 001 |
| Paper | 010 |
| Scissor | 011 |
| Lizard | 100 |
| Spock | 101 |



Then game compares player's choice with computer's choice and selects winner of the round according to Table1. If there is a tie, no one gets marks.
Updated marks are displayed by corresponding LED sets.

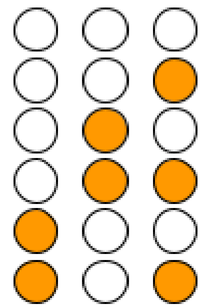
**Computer Marks**

| | |
|---|---|
| Mark = 0 | 000 |
| Mark = 1 | 001 |
| Mark = 2 | 010 |
| Mark = 3 | 011 |
| Mark = 4 | 100 |
| Mark = 5 | 101 |



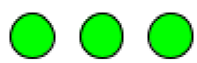
**Player Marks**

| | |
|---|---|
| Mark = 0 | 000 |
| Mark = 1 | 001 |
| Mark = 2 | 010 |
| Mark = 3 | 011 |
| Mark = 4 | 100 |
| Mark = 5 | 101 |

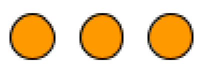


Next, within few seconds the next round starts.
This repeats until 7 rounds are completed.
When all rounds are completed, the player who has the highest marks is considered as the winner.

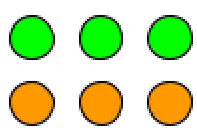
If the winner is computer, LEDs corresponding to computer illuminates.



If the winner is player, LEDs corresponding to player illuminates.



In case of a draw, all LEDs corresponding to computer and player illuminates.



As an option to display the winner in the above way, the total marks of the players can be displayed as binary.
Then, buzzer buzzes to indicate the end of the game and LEDs corresponding to the computer's choices will blink to mark the end.

★ The game can be ended by pressing the start/end button anytime.